CREDIT CARD FRAUD DETECTION (FRAUD SHIELD AI)

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PROBLEM STATEMENT & OVERVIEW



Building a Smarter Defense Against Financial Fraud

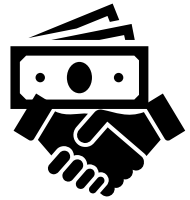
- Fraud is growing with digital transactions and advanced tactics. Traditional rules miss new fraud patterns
  - We use **machine learning and deep learning** for better fraud detection.
- The goal of this project is to build a **scalable fraud detection system** that finds known and emerging fraud while reducing false positives and improving financial security.



Detect Fraud Pattern



Reduce False Positives



Strengthen Financial Security & Trust

DATASET OVERVIEW

Source: Kaggle – Credit Card Transaction Fraud Detection

Dataset Size: ~1.3 million Credit Card Transactions

Nature: Realistic Simulation of Fraudulent and Legitimate transactions

Key Features

Transaction: Date/Time, amount, category

Merchant: Merchant Name and Geographic coordinates

Cardholder: Card number, location, city population

Target: Fraud indicator (is_fraud: 1 = Fraud, Is_fraud: 0 = Legitimate)

EXPLORATORY DATA ANALYSIS

Understanding the data to ensure quality and identify key patterns

1. DATA QUALITY

1.3 M Transactions

0
Missing Transactions

0
Duplicate Records

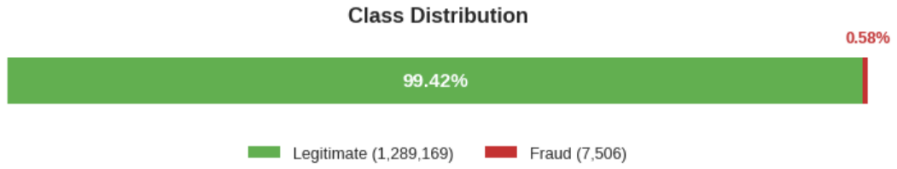
Data Types Standardized
cc num, Zip (String), DOB (Date Time)

Outliers Identified
Transaction Amount & City Population

2. CLASS DISTRIBUTION

Highly Imbalanced Dataset

Fraud represents only 0.58% of all transactions



Class Imbalance will be handled using advanced techniques like SMOTE during model training

3. KEY EDA INSIGHTS

Average Transaction Amount
$70.35

Median Transaction Amount
$47.52

Maximum Transaction Amount
$28,948.90

Fraud Rate
0.58% (7,508 transactions)

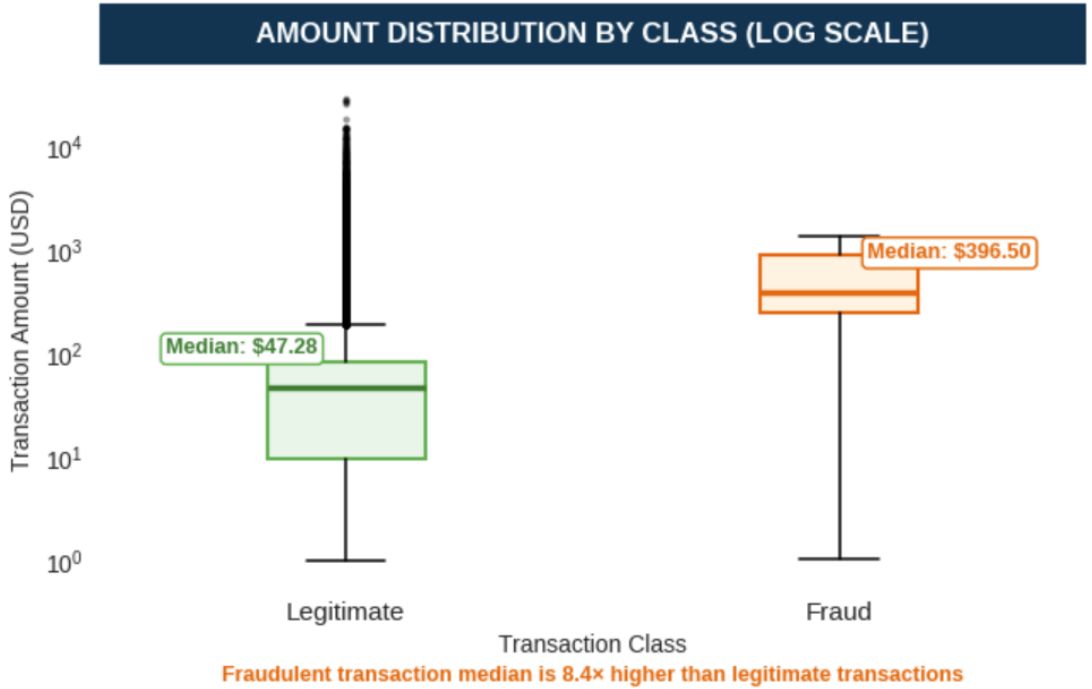
Key Finding: Dataset is clean with no missing values or duplicates but highly imbalanced

TRANSACTION AMOUNT ANALYSIS

Fraudulent transaction amounts are significantly higher, with a median of **$396.50** vs. **$47.28** for legitimate transactions—approximately **8.4×** higher.

Total Amount (Legit)
$87,234,340.29

Total Amount (Fraud)
$3,988,088.61

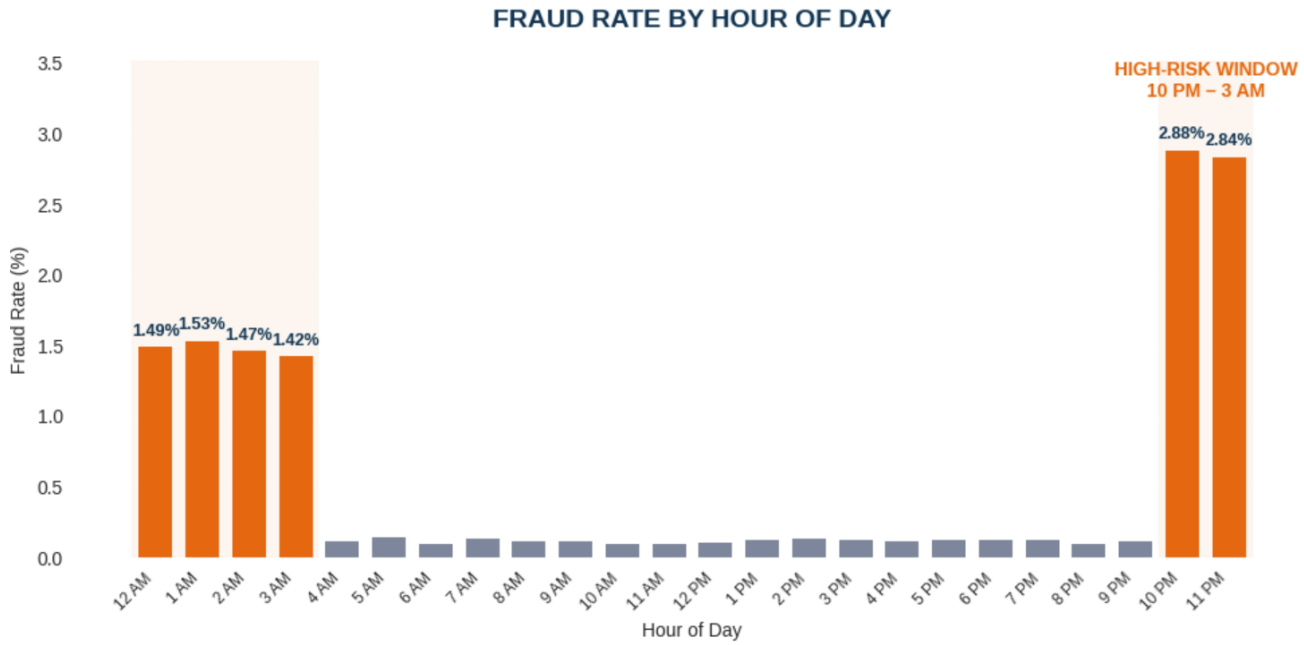


AMOUNT STATISTICS BY CLASS					
CLASS	MEAN	MEDIAN	STD	MIN	MAX
Legit (0)	$67.67	$47.28	$154.01	$1.00	$28,948.90
Fraud (1)	$531.32	$396.50	$390.56	$1.06	$1376.04

Key Insight: Fraudulent transactions have significantly higher amounts across most summary statistics

TIME PATTERN ANALYSIS

Fraud is highly concentrated at night, with a **2.087%** fraud rate from **10 PM – 3 AM , 18.09X** higher than day time.



PEAK RISK HOURS	10 PM	11 PM
	2.88%	2.84%
	1,931 Fraud Trans	1,904 Fraud Trans

NIGHT | 10 PM – 3 AM

2.087% Fraud Rate	304,882 Transactions
	6,362 Fraud cases

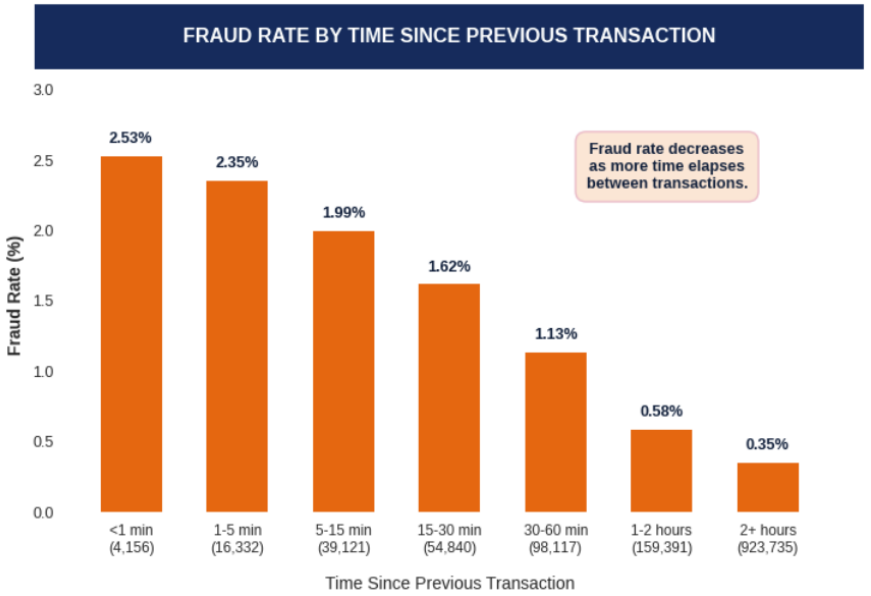
18.09X
Higher Fraud Rate

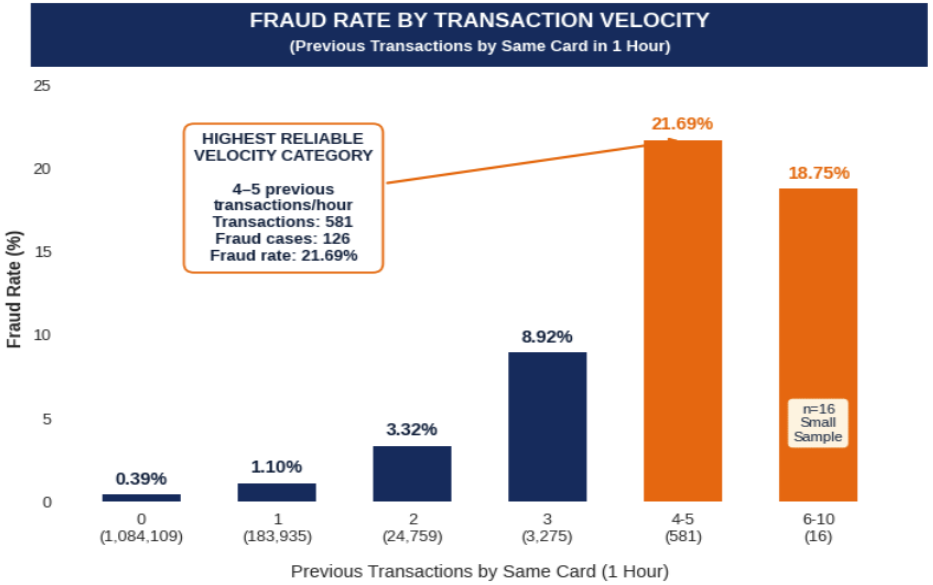
DAY | 4 AM – 9 PM

0.115% Fraud Rate	991,793 Transactions
	1,144 Fraud cases

TRANSACTION VELOCITY ANALYSIS

Fraud risk rises sharply with repeated card usage: transactions with **4–5 prior transactions in one hour** show a **21.69% fraud rate vs. 0.39%** with no prior transactions — a **55.45× increase**.





21.69%
HIGH-VELOCITY FRAUD RATE
(4-5 Prior Transaction/hour)
581 transactions | 126 fraud

0.39%
BASELINE FRAUD RATE
(0 Prior Transaction/hour)
1.1 M transactions | 4,240 fraud

55.45 x
RELATIVE RISK INCREASE
(21.69% VS 0.39%)
Higher observed fraud rate

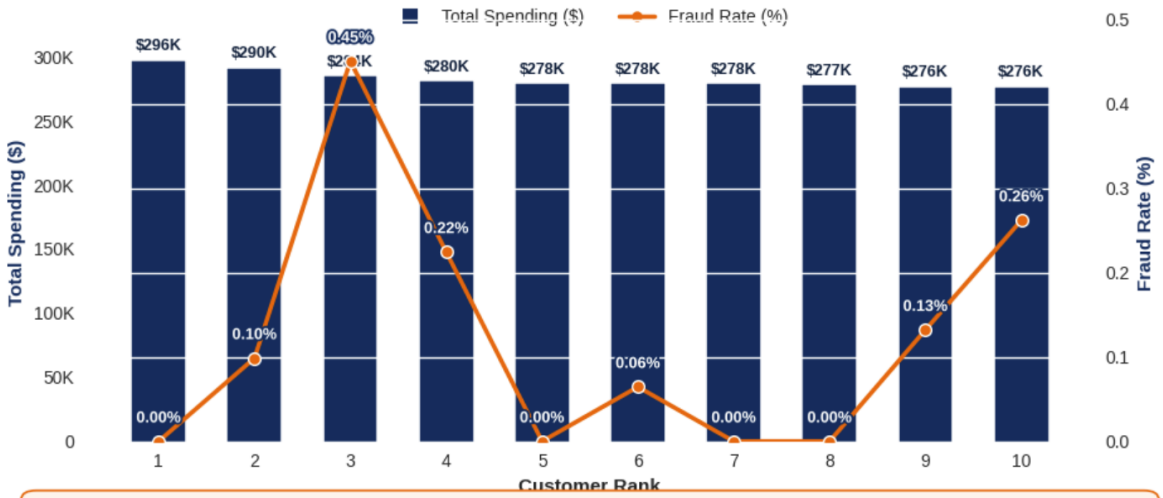
CUSTOMER SPENDING HABIT ANALYSIS

SPENDING CONCENTRATION
27.4% of customers account for 50% of total spending

TOP 10 CUSTOMER FRAUD RATE
0.124%
(vs 0.579% overall fraud rate)

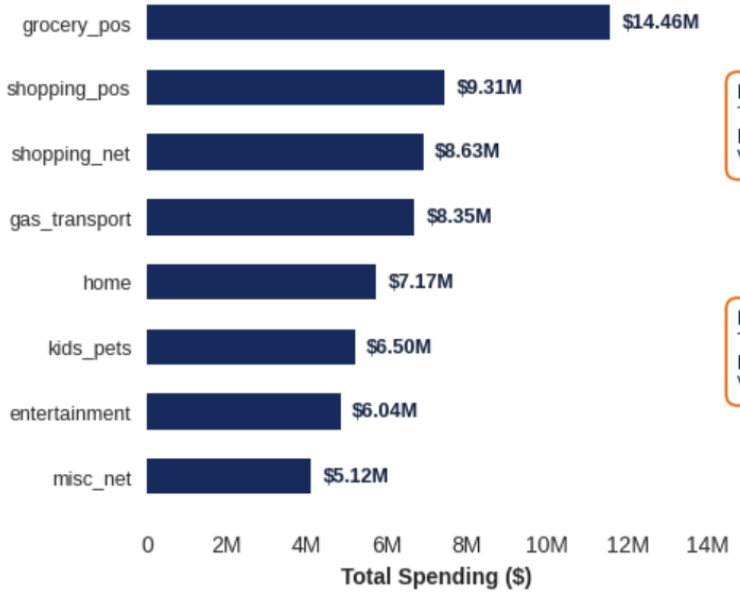
High-spending customers show low fraud risk (0.124%), while 27.4% of customers drive 50% of total spending.

TOP 10 CUSTOMERS BY TOTAL SPENDING



Top 10 Spending Customers	Fraud Rate	Overall Fraud Rate	Relative to Overall
30,736 transactions | 38 fraud cases	0.124%	0.579%	0.21x lower

TOTAL SPENDING BY MERCHANT CATEGORY



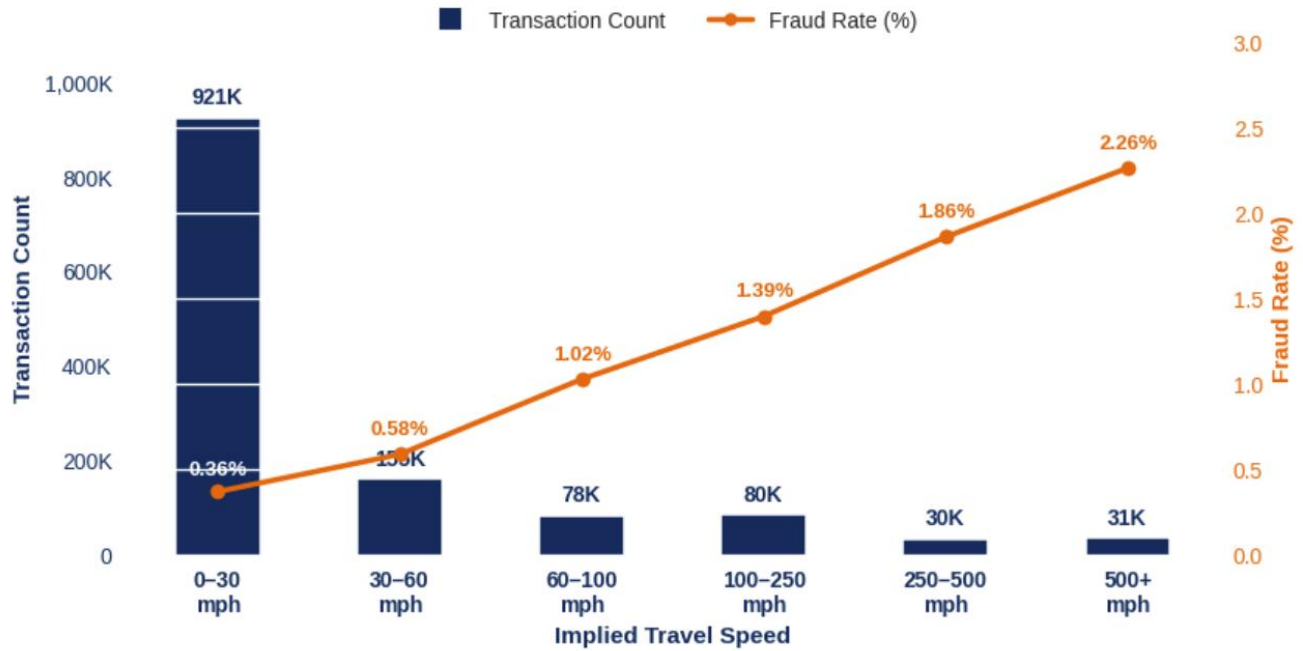
KEY CATEGORY — grocery_pos
Total Spending: $14.46M
Fraud Rate: 1.41%
Vs Overall: 2.44×

KEY CATEGORY — shopping_net
Total Spending: $8.63M
Fraud Rate: 1.76%
Vs Overall: 3.03×

KEY TAKEAWAY: High spending alone is not a strong fraud indicator; top spenders had a 0.124% fraud rate vs. 0.579% overall. Combining customer spending patterns, merchant category, transaction amount, velocity, and timing provides stronger signals for identifying high-risk transactions.

GEO SPATIAL METRIC ANALYSIS

Implied travel speed estimates how fast a cardholder would need to travel between two consecutive transaction locations, based on the distance traveled and time between transactions. Unusually high speeds can indicate geographically impossible or suspicious card activity.



Key Insight: Transactions in the **500+ mph** category have a **2.258% fraud rate**, **~3.90X** the overall fraud rate of **0.579%.** This suggests unusually rapid geographic movement between consecutive transactions may be a valuable fraud-detection feature.

EDA INSIGHTS → FEATURE ENGINEERING

The Fraud model should **prioritize behavioral and transaction-level features**

- Rapid repeated transactions on the same card
- Late night transactions (10 PM – 3 AM)
- Larger than normal transaction amounts
- Certain Merchant Categories
- Unusually high Implied Travel Speeds between consecutive transactions

High Value Model Features

- Amt (Transaction Amount)
- Hour Of Day
- Category
- Transaction Velocity
- Minutes & Distance Since Previous Transaction
- Implied Travel Speed

MODEL BUILDING & EVALUATION

Comparing multiple machine learning models to find the best fraud detection system

Goal:
Identify the model with the best balance of high recall and low false positives

Dataset:
~ 1.3M transactions
Fraud Rate : ~0.58%

Evaluation:
Test set with validation tuned threshold

Metric:
Recall, F1-Score, Precision, PR-AUC

1. **Baseline Models:** Logistic Regression, Decision Tree, KNN
2. **Ensemble Models:** Random Forest
3. **Boosting Models:** XGBoost, LightGBM
4. **Neural Network:**
5. **Threshold Tuning:** Optimize threshold on validation set and evaluate on test set

HELD-OUT TEST LEADERBOARD

Validation-Tuned Threshold

Rank	Model (Threshold)	Precision	Recall	F1	ROC-AUC	PR-AUC	Fit Time (sec)
1	XGBoost @ 0.95 (test)	0.76	0.80	0.78	1.00	0.86	34.5
2	Neural Network @ 0.30 (test)	0.87	0.68	0.77	1.00	0.82	60.3
3	Random Forest @ 0.85 (test)	0.78	0.73	0.75	0.99	0.80	126.5
4	Decision Tree @ 0.95 (test)	0.40	0.82	0.53	0.97	0.56	35.0
5	LightGBM @ 0.95 (test)	0.47	0.78	0.59	0.97	0.44	25.4
6	KNN @ 0.15 (test)	0.62	0.41	0.50	0.80	0.44	0.0
7	Logistic Regression @ 0.90 (test)	0.30	0.56	0.39	0.96	0.22	15.6

Best metric values highlighted in green.

Final Model Selection : XG Boost delivers the best overall balance with 80% recall and 76% precision

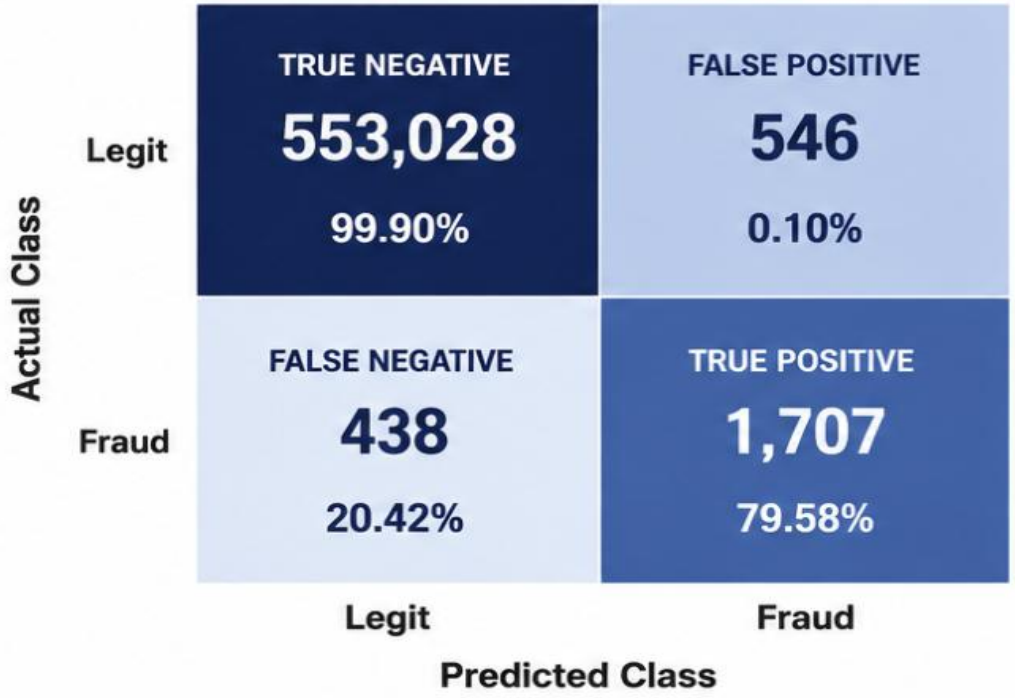
XG BOOST MODEL

PERFORMANCE METRICS

METRIC	WHAT IT MEANS	XGBOOST
Precision	When the model predicts fraud, how often it is right	75.77%
Recall	How well the model finds all actual fraud cases	79.58%
F1 Score	Balance between precision and recall	77.63%
PR-AUC	Overall ability to identify fraud across thresholds	86.09%

XGBoost Confusion Matrix

Validation-tuned threshold: 0.95



KEY TAKEAWAY: XGBoost detects **79.6%** of actual fraud while **75.8%** of fraud alerts are correct, achieving a PR-AUC of **86.1%**.

RECOMMENDATION

Cost – Benefit Summary : XG Boost detected **79.6%** of fraud transactions and captured **78.9%** of fraud value (**$120,784 of $153,150**), reducing estimated fraud-related cost by **31.2%** and delivering **$47.7K** in net savings at a **65.3% ROI**.

Fraud Recall (Transaction)
79.6%
1,707 of 2,145

Fraud Value Captured
78.9%
$120,784 of $153,150

Estimated Net Savings
$47.7K
vs. no fraud model

Cost Reduction
31.2%
Reduction in fraud related cost

ROI
65.3%
Benefit / Operating Cost

1. COST COMPARISON

Estimated Fraud Cost



Estimated savings of $47,734 (31.2% reduction in fraud-related cost)

2. FRAUD VALUE BREAKDOWN

Fraud Dollar Value



XGBoost captured $120,784 of total $153,150 fraud value.

3. COST COMPONENTS (WITH XGBOOST)

Breakdown of Total Cost ($105,415)



Total operational cost (FP + Investigations): $73,050

KEY BUSINESS IMPACT

Lower Fraud Loss
Reduced remaining fraud loss to $22,365 from $153,150 without a model.

Efficient Operations
2,253 alerts reviewed with high precision (75.8%) to minimize disruption.

Strong Financial Return
For every $1 spent on operations, $1.65 of fraud value is captured.

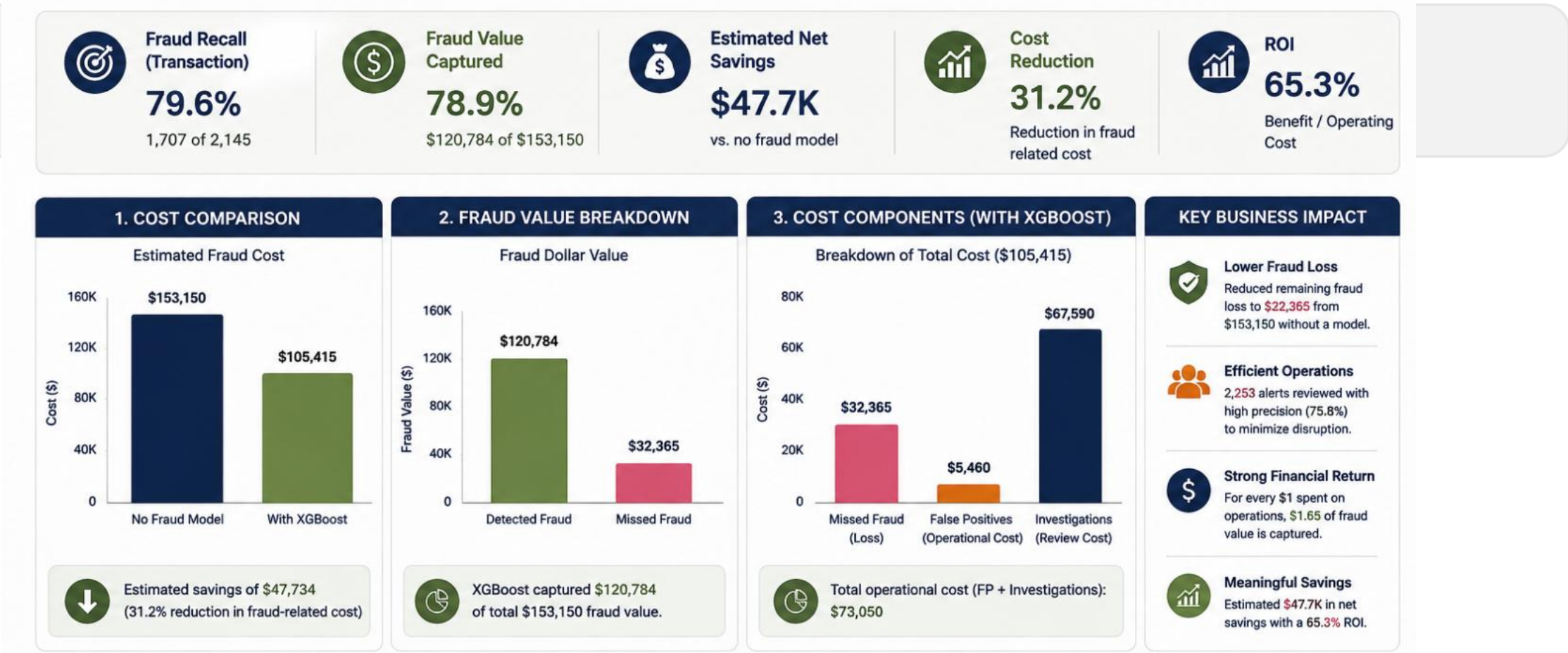
Meaningful Savings
Estimated $47.7K in net savings with a 65.3% ROI.

THANK YOU

QUESTIONS?

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XG BOOST MODEL

Model Performance

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